Mathematical model

Page 1 - vector mathematics and selectable text.

This small fixture checks faithful typesetting, page order, and reading flow. Each page contains text and vector diagrams, without image-based equations.

A variational model

Let $u \in H_0^1(\Omega)$ minimize the following functional:

$$\mathcal{E}[u] = \int_\Omega \left(\frac{1}{2} \|\nabla u\|^2 + \frac{\lambda}{4} u^4 - f u \right) \mathrm{d}x.$$

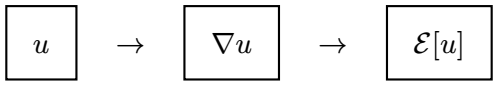
For a test function v, the first variation satisfies

$$\frac{d}{d\varepsilon} \mathcal{E}[u + \varepsilon v] \mid_{\varepsilon = 0} = \int_\Omega \left(\nabla u \cdot \nabla v + \lambda u^3 v - f v \right) \mathrm{d}x = 0.$$

The corresponding spectral approximation uses

$$u_{N(x)} = \sum_{k=1}^{N} a_k \varphi_{k(x)}, \quad \boldsymbol{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

A vector diagram



The original PDF retains its fonts, margins, equations, and vector paths.